

Yingzi Ma

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RESEARCH INTERESTS

I am a second-year PhD student majoring in Information Science at the University of Wisconsin-Madison, advised by Chaowei Xiao. Now, I am collaborating with Yulong Cao, an NVIDIA researcher at the NVIDIA Autonomous Vehicle Research group. My research interests are:

- Multi-modal Large Language Models (MLLMs).
- End-to-end Autonomous Driving System.
- Parameter-efficient fine-tuning (PEFT) of large-scale Pre-trained Language Models (PLMs).

EDUCATION

University of Wisconsin Madison, Madison, U.S. Sept. 2024 | Jun. 2029 (expected)
PhD in Information Science Cumulative GPA: 3.85/4.00

Sichuan University, Chengdu, China Sept. 2020 | Jun. 2024 (expected)
Bachelor of Engineering (BEng) in Artificial Intelligence Cumulative GPA: 3.84/4.00
China National Scholarship, 2021 Score Point Average: 91.26 (Rank, 3/39)

Sichuan University, Chengdu, China Sept. 2019 | Jun. 2020
Bachelor of Arts (BA) in Tourism Management Cumulative GPA: 3.86/4.00
China National Scholarship, 2020 Score Point Average: 90.67 (Rank, 1/102)

PUBLICATIONS

Conference Papers

- **Dolphins: Multimodal Language Model for Driving** [[AD](#), [MLLM](#)]
Yingzi Ma, Yulong Cao, Jiachen Sun, Marco Pavone, Chaowei Xiao
Published on ECCV 2024. [WebPage][PDF]
- **Benchmarking Vision Language Model Unlearning via Fictitious Facial Identity Dataset** [[MLLM](#)]
Yingzi Ma, Jiong Xiao Wang, Fei Wang, Siyuan Ma, Jiazhaoli, Xiujuan Li, Furong Huang, Lichao Sun, Bo Li, Yejin Choi, Muhao Chen, Chaowei Xiao
Published on ICLR 2025.
- **dVLM-AD: Enhance Diffusion Vision-Language-Model for Driving via Controllable Reasoning** [[AD](#)]
Yingzi Ma, Yulong Cao, Wenhao Ding, Shuibai Zhang, Yan Wang, Boris Ivanovic, Ming Jiang, Marco Pavone, Chaowei Xiao
Submit to CVPR 2026. [PDF]
- **SafeVL: Driving Safety Evaluation via Meticulous Reasoning in Vision Language Models** [[AD](#)]
Yingzi Ma, Yulong Cao, Wenhao Ding, Yuxiao Chen, Xinshuo Weng, Yan Wang, Wenjie Luo, Yurong You, Boyi Li, Yue Wang, Rachel Luo, Sushant Veer, Xiaopeng Li, Approva Sharma, Ed Schmerling, Boris Ivanovic, Marco Pavone, Chaowei Xiao
Submit to ICRA 2026. [PDF]
- **ARMOR: Aligning Secure and Safe Large Language Models via Meticulous Reasoning** [[LLM](#)]
Zhengyue Zhao, Yingzi Ma, Somesh Jha, Marco Pavone, Chaowei Xiao
Arxiv. 2025. [PDF]
- **MIXLORA: Enhancing Large Language Models Fine-Tuning with LoRA-based Mixture of Experts** [[PEFT](#)]
Dengchun Li, Yingzi Ma, Naizheng Wang, Zhengmao Ye, Zhiyuan Cheng, Yinghao Tang, Yan Zhang, Lei Duan, Jie Zuo, Cal Yang, Mingjie Tang
Arxiv. 2024. [PDF]
- **MULTI: Multimodal Understanding Leaderboard with Text and Images** [[MLLM](#)]
Zichen Zhu, Yang Xu, Lu Chen, Jingkai Yang, Yichuan Ma, Yiming Sun, Hailin Wen, Jiaqi Liu, Jinyu Cai, Yingzi Ma, Situo Zhang, Zihan Zhao, Liangtai Sun, Kai Yu
Arxiv. 2024. [PDF]

- **Understanding the Multimodal Instruction Format for In-Context Learning** [MLLM]
Yingzi Ma, Chunyuan Li, Chaowei Xiao
Preprint. 2023.
- **CAM-GUI: A Conversational Assistant on Mobile GUI** [MLLM]
Zichen Zhu, Liangtai Sun, Jingkai Yang, Yifan Peng, Wutao Li, Weilin Zou, Ziyuan Li, Lu Chen, Yingzi Ma, Danyang Zhang and Kai Yu
Published on *NCMMSC*. 2023. [PDF]
- **SK-Adapter: Syntax Enhanced Pre-trained Language Models with Structured Knowledge Adapter** [PEFT]
Yingzi Ma, Da Ma, Zichen Zhu, Yunyan Zhang, Ruihui Zhao, Huimin Wang, Yefeng Zheng and Lu Chen
Preprint. 2023

PROFESSIONAL SERVICES

- ICLR 2024 Reviewer
- ICLR 2025 Reviewer
- ICCV 2025 Reviewer
- TMLR 2025 Reviewer
- ICRA 2026 Reviewer
- ICLR 2026 Reviewer

ACADEMIC EXPERIENCE

University of Wisconsin, Madison (UWM)

Research Intern

Madison, WI 53706, United States

Apr. 2023 | Present

- Working with Prof. Chaowei Xiao at University of Wisconsin–Madison and Chunyuan Li, a principal researcher at Microsoft Research, Redmond, we Exploring the effects of different data formats on multimodal in-context learning.
- Collaborating with Prof. Chaowei Xiao and some researchers from NVIDIA, we aim to extend the powerful capabilities of MLLMs to autonomous driving-related videos.
- All tasks, including coding and conducting experiments, were carried out solely by me in these two research projects.

AWS Nova AI Challenge 2025

Red Team Member

Seattle, United States

December. 2024 | July. 2025

- Participated in the AWS Nova AI Challenge 2025 as a red team member.
- Implemented and trained utterance reward models to perform test-time scaling attacks.
- Used PPO (Proximal Policy Optimization) to train attacker models that prompt defended models into generating vulnerable code.

Shanghai Jiao Tong University (SJTU)

Research Intern

Shanghai, China

May. 2022 | Apr. 2023

- Work as a research intern advised by Prof. Lu Chen in the language model group of the Cross Media Language Intelligence Lab (X-LANCE Lab) at SJTU.
- Contribute to enhancing the performance of PLMs on NLU and NLG tasks by injecting syntax structure information through structured knowledge adapters (SK-Adapter).
- Collaborate with researchers from Tencent Jarvis Lab to apply PLMs on the Chinese medical corpus.

Sichuan University (SCU)

Team Member

Chengdu, China

Sept. 2021 | Sept. 2022

- Collaborate with four other members of a team to carry out a project of **CBCT jaw region segmentation method based on context fusion mechanism**.
- Attend the National College Students' Innovation and Entrepreneurship Training Program, 2022, and receive the provincial-level assessment.
- Publish a patent for an invention as the first inventor (**CN115578406A**).

SKILLS

- **Languages:** Mandarin, English, Japanese
- **Programming:** Python, C/C++, $\text{\LaTeX}$, MATLAB, Mathematica, Javascript
- **Machine Learning:** PyTorch, TensorFlow, Sklearn, PaddlePaddle, Stable_baselines3

AWARDS

- **Finalist**, Mathematical Competition In Modeling (MCM), 2022
- **Honorable Mention**, Interdisciplinary Contest In Modeling (ICM), 2021
- **2nd Award**, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM), 2021
- **2nd Award**, Chinese Mathematics Competitions (CMC), 2020